Loong Yi **Lee**

李隆亿, Robotics Researcher

Helping people & robots reach their goals, together.

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Scholar Website

Education

Ph.D. in Mechanical Engineering **2018 – 2022**

Monash University, Malaysia

Thesis: *Design, Optimization, and Development of Soft Robotic Grippers for Aerial Grasping*
Supervisors: A./Prof. Surya G. Nurzaman & Prof. Chee Pin Tan

B.Eng. (Hons) in Mechatronics Engineering **2014 – 2018**

Monash University, Malaysia

Thesis: *Development of a Mobile Platform for Collaborative Robots* (First Class Honours)

Research Interests

Soft Robotics Embodied Intelligence Robot Manipulation Design Optimization Human-Robot Interaction

Research Experience

Senior Research Fellow **April 2025 – 2026**

Monash University, Malaysia

Supervisor / Manager: A./Prof. Surya G. Nurzaman

- Initiated project: Novel robot systems for agile locomotion in uncertain terrain via morphological intelligence.
- Joined an ongoing project on contact-rich aerial robotic inspection via shape-shifting drones, funded by Petronas.

Research Associate **September 2022 – March 2025**

University of Bristol & the Bristol Robotics Laboratory, UK

Supervisor / Manager: Prof. Jonathan Rossiter & Prof. Nathan Lepora

- Project: Autonomous dexterous manipulation (soft robots & tactile sensors) with the MSI Smart Cobotics Centre.
- Investigated novel soft gripper designs with morphologically intelligent design paradigms for embodied dexterity.
- Deployed deep-learning based tactile sensing solutions for fine manipulation problems, including tool use.
- Investigated novel electro-adhesion, and dry-adhesion mechanisms for robot manipulation.
- Investigated suction and fluidics as a medium for embodied intelligence in soft robots.
- Developed a bimanual manipulation testbed, complete with 3D vision, force-torque sensing, tele-operation for robot learning, ROS2 and RL policy rollout readiness, demonstrated via autonomous bin-picking of food items.
- Investigated how users can trust soft robots via a trustworthiness design metric, tested across a manipulation design space via a modular soft robot manipulator system. As part of the UKRI TAS Functionality Node.
- Demonstrated project output at various public events and engaged with industry partners in collaborative research.

Doctoral Student **July 2018 - April 2022**

Monash University, Malaysia

Supervisor / Manager: A./Prof. Surya G. Nurzaman & Prof. Chee Pin Tan

- Project: Design, Optimization, and Development of Soft Robotic Grippers for Aerial Grasping
- Developed platforms which were used to identify challenges in mobile manipulation.
- Designed aerial soft grippers that are robust to size, positional, and orientation variances for aerial pick-and-place.
- Developed and introduced rapid reality-assisted data-driven co-optimization pipelines for soft grippers.

Research Assistant **July 2018 - December 2018**

Monash University, Malaysia

- Project: Soft Service Robots to assist older adults in everyday living.
- Developed a mobile manipulator platform with a soft gripper to pick up household objects. Evaluated perception of the robot system by older adults via quantitative surveys with an adapted technology readiness model.

Higher Education Teaching Experience

Dissertation supervisor & examiner **2022 – 2025**

School of Engineering Mathematics & Technology, University of Bristol

- Proposed, supervised, and examined dissertation projects for the MSc. Robotics, and MSc. Biorobotics programs.
- Supervision led to students producing three peer-reviewed publications and one cohort best project award.

Tutor / lab demonstrator **2020 – 2021**

Mechatronics Engineering, Monash University Malaysia

- TRC4802 "Thermo-fluids & power systems" - Developed, delivered group problem-solving tutorial sessions.
- TRC4901 "Artificial intelligence for engineers" - Demonstrated, assessed practical lab activities for remote-learning.

Peer-assisted study session leader **2019 – 2020**

Mechanical Engineering, Monash University Malaysia

- MEC2402 "Engineering Design 1" - Mentor for student design project groups.
- Coached the 2019 student design team towards the nation's first silver medal at the WARMAN Engineering Design Challenge international final in Sydney.
- Assisted with the development of & facilitated remote-learning based student design projects.

Non-University Teaching Experience

Part-time freelance school robotics coaching **2013 – 2022**

Schools: SMK Seafield, SMK USJ 13, KLCR Centre Hope Academy, Sri KL International School

- Coached student teams preparing for robotics competitions such as FLL and WRO.
- Developed workshops & courses tailored for students' current STEM curriculum progress to bridge into robotics.
- Delivered a curriculum for a special needs program to introduce neurodivergent students to robotics.

Master Trainer **2013 – 2014**

Creative Robotics, One City Learning Center

- Helped start a new learning center for robotics education for ages 6-17.
- Delivered a structured curriculum with LEGO Mindstorms robot kits and monitored student progress.
- Coached student teams to success in national and international robotics competitions.

Industry Experience & Engagement

Industry-Collaboration Project **2025-2026**

Aerial robotic inspection, Petronas and Monash

- Routine reporting with project stakeholders on project progress and plans.

Industry Feasibility Project **2024-2025**

Tactile paste manipulation, via the Smart Cobotics Centre

- Co-developed and led a sprint feasibility project with four industry partners, raising 20,000 GBP of investment.
- Successfully completed the 6-month feasibility project with demonstrations at public events.

Industry-Collaborative Research **2022-2025**

Soft & dexterous manipulation, Ocado and Bristol

- Delivered regular research updates and fostered discussions as part of a long term research collaboration.

Industry internship **November 2016 – February 2017**

Robotics & Motion division, ABB Malaysia

- Developed simulated robot demos in Robotstudio for client feasibility checks and visualisation.
- Developed and presented robot demonstrations at tradeshows and festive videos with the YuMi Cobot.
- Assisted project & service teams with onsite operations.
- Proposed a collaborative project based on my experience there which formed by honours thesis.

Service & Community Engagement

Service at The 9th IEEE-RAS International Conference on Soft Robotics (RoboSoft 2026) April 2026

- On-site representative for Exhibition Co-chair. (Representing A./Prof. Surya Nurzaman)
- Best Extended Abstract Award committee member.

Examiner for Final Year Projects, Mechanical Engineering, Monash University Malaysia 2025-2026

New Lab Startup, Soft Materials Robotics & Technologies (SMART) Lab, Monash University Malaysia 2025

- Managed transition to a new lab space, procurement of equipment and consumables, personnel management & training. Introduced new technology processes and developed robot manipulation and experiment platforms.

Co-organizer, Bristol Softlab Research Seminars 2023-2025

- Engaged experts to present and discuss their work to members of the Bristol Robotics Laboratory.

Mentor, Shell Eco-marathon Team Monash 2019-2022

- Mentor for a student team that develops fuel-efficient hydrogen fuel cell vehicles, with an award-winning autonomous vehicle initiative. Previously held the role of team manager, managing a team of 50 members.

Review work

- Peer reviewed for THRI, T-Mech, RAM, RAL, IROS, IEEE TIE, Frontiers, Robosoft, IOP trusted peer reviewer.

Selected Peer-Reviewed Publications

L. Y. Lee, S. Terrile., S. Nam., T. Liang, N. Lepora, & J. Rossiter, Fin-A-Rays: Extending Soft Gripper Compliance via Discrete Arrays of Flexible Structures, *Soft Robotics*, *13*(1), 121–135, 2025.

L. Y. Lee, O. A. Syadiqeen, Z. H. Chin, & S. G. Nurzaman, Customizable Power Grasps From Uncrewed Aerial Vehicles via a Soft 'Wrapping' Gripper, *IEEE/ASME Transactions on Mechatronics*, *29*(2), 972-983, 2024.

L. Y. Lee, O. A. Syadiqeen, C. P. Tan, & S. G. Nurzaman, Closed-Structure Compliant Gripper With Morphologically Optimized Multi-Material Fingertips for Aerial Grasping, *IEEE Robotics and Automation Letters*, *6*(2), 887–894, 2021.

S. A. Katiyar*, **L. Y. Lee***, F. Iida, & S. G. Nurzaman, Energy Harvesting for Robots with Adaptive Morphology, *Soft Robotics*, *10*(2), 265–279. 2023. *co-first author

L. Y. Lee, W. M. Lim, P.-L. Teh, O.A. Syadiqeen, & S. G. Nurzaman, Understanding the Interaction between Older Adults and Soft Service Robots: Insights from Robotics and the Technology Acceptance Model, *AIS Transactions on Human-Computer Interaction*, *12*(3), 125–145. 2020.

L. Y. Lee, & S. G. Nurzaman, A Gripper for Compliant Grasping of an Object, Patent no. PI2021006636 (MY), IDP000099149 (ID), both granted in 2025.

T. Yue., C. Lu, K. Tang., Q. Qi, Z. Lu, **L. Y. Lee**, H. Bloomfield-Gadêlha, J. Rossiter, Embodying Soft Robots with Octopus-inspired Hierarchical Suction Intelligence, *Science Robotics*, *10*(102), eadr4264. 2025.

A. Hicks, **L. Y. Lee**, R.S. Diteesawat, A. Roshan, Y. Zekaria, E. Hall, E. Werner, N.Rahman, E. Czech, J. Rossiter, Soft Robotic Technological Probe for Speculative Fashion Futures, *ACM Transactions on Human-Robot Interaction*, *5*(3), 1-28. 2026.

S. Nam, B. Deng, **L. Y. Lee**, J. Rossiter, & N. Lepora, TacFinRay: Soft Tactile Fin-Ray Finger with Indirect Tactile Sensing for Application to Robust Grasping, *IEEE Robotics and Automation Letters*, *11*(3), 2722–2729. 2026.

L. Tan, **L. Y. Lee**, A. Kulykov, J. Rossiter, & A. T. Conn, Gecko-inspired adhesion enhanced by electroadhesive forces, *Smart Materials & Structures*, *33*(12), 125022. 2024.

Selected Peer-Reviewed Conference Presentations

L. Y. Lee, S. Terrile, A. Roshan, T. Yue, S. Nam, & J. Rossiter, Down the Rabbit Hole: Exploiting Airflow Interactions for Morphologically Intelligent Retracting Vacuum Grippers, *Robosoft 2024 Poster*.

C. Liu, **L. Y. Lee**, S. Terrile, S. Nam, N. Lepora, & J. Rossiter, Feeling the Pinch: Differentiating Thin Materials with a Soft Vision-based Tactile Sensor, *Robosoft 2026 Oral*.

R. Wu, **L. Y. Lee**, S. Terrile, & H. Hauser, Physical Learning in Soft Fluidic Channels via Experience, *Robosoft 2026 Oral*.

T. Y. M. Sim, **L. Y. Lee**, & S. G.Nurzaman, Bayesian Optimization of Pneumatic Soft Grippers via Reconfigurable Modular Molds, *Robosoft 2023 Oral*.

Alternative Research Contributions & Presentations

Invited talk, in-person seminar at School of Data Science, Kyungpook National University, S. Korea, January 2026
- Presentation title: 'Finding Function from Structure: Design & Optimization of Soft Robotic Manipulation'.

White paper contribution, The Future of Smart Human-Robot Collaborative Working Environments: Towards Meaningful Work for All: A White Paper, 2025

Poster Contributions, UK Manipulation Workshop
- King's College London 2025, Oxford University 2024, Bristol Robotics Laboratory 2023.

Poster contribution, ICRA 2023 ViTac Workshop. **Breakout session talk**, Embodied Intelligence Conference 2024

Public Robot Demonstrations: ITEX Malaysia 2021, UK Smart Manufacturing Week 2024, UK-RAS Festival of Robotics 2024, MTC Robotics & Automation event 2024, UKRI TAS Showcase 2025, Innovate UK Materials & Manufacturing Showcase 2025, MSI Smart Cobotics Open Day 2024, UK Manipulation Workshop 2023.

Funding, Awards, & Honours

[Honour] Joint-Appointed Professorship program, Kyungpook National University, S. Korea	2026 – present
[Fund] IEEE RAS Member Support Program Travel Grant for Robosoft 2026, 1500 USD.	2026
[Fund] Global Matching Lab startup fund, 3500 USD, Kyungpook National University	2026
[Honour] Honorary Research Associate, University of Bristol.	2025 – present
[Award] Best design award as team leader, Robosoft 2024 competition.	2024
[Award] Winner as team technical lead, Functional Fashion competition at IROS 2023.	2023
[Award] Best oral presentation award finalist, Robosoft 2023.	2023
[Award] Best locomotion award as team leader, Robosoft 2023 locomotion competition.	2023
[Award] Gold Medal at the International Invention, Innovation, Technology Exhibition (ITEX) 2021.	2021
[Funding] Monash Graduate Research Excellence Scholarship. Tuition and stipend coverage.	2021 – 2022

Finalist in DJI Robomaster AI Challenge, ICRA 2018, awarded a 7500 USD robot kit & 1000 USD travel sponsorship.

Unsucessful bids for British Council ISPF IREP International collaboration grant, ARIA new dexterity program, and Leverhulme Trust research grant.

Relevant Skills / Training

Software: Solidworks-CSWA certified, Autodesk Inventor/Fusion, Simulation (ANSYS, MuJoCo, Isaac Sim), Programming (C, C++, Python, MATLAB, Scratch), ROS/ROS2, PyTorch (DL, RL), OpenCV, PLC.

Hardware: Soft robot fabrication, additive manufacturing / 3D printing, robot arms, embedded programming, instrumentation & data acquisition, robot tele-operation & behavior cloning, robot actuators (Dynamixel, FOC drives).

Languages: English, Bahasa Malaysia, Mandarin.

Professional Affiliations

Institute of Electrical & Electronic Engineers (IEEE), member	2018 – Present
IEEE Robotics & Automation Society (RAS), member	2018 – Present
Association for Computing Machinery (ACM), member	2026 – Present
Board of Engineers Malaysia (BEM), graduate member	2020 – Present

Referees

A./Prof. Surya G. Nurzaman
LIFE Lab
Associate Professor,
Monash University Malaysia
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Prof. Jonathan Rossiter
Bristol SoftLab
Professor of Robotics,
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Prof. Nathan Lepora
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